

UPI Transaction Analysis

Data-Driven Risk & Performance Intelligence

- Data Analytics with GenAI · Capstone Project
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Why this analysis



Growth

100K transactions · ₹42.4L value



Fraud Risk

2.0% fraud · 2,000 alerts



Reliability

5.87% failure rate

OBJECTIVES

01

Analyze transaction behaviour

02

Identify fraud & failure drivers

03

Develop executive KPIs

04

Recommend actions

One question drives the deck: where does risk actually live in a 100K-transaction UPI platform?

The data foundation

upi_transaction_history 100,000

device_info 12,000

upi_account_details 12,000

customer_master 10,000

customer_feedback_surveys 4,000

fraud_alert_history 2,000

merchant_info 500

PIPELINE

Excel



SQL



Python



Statistics



Power BI

7 tables · 138,500 records audited for quality, keys and schema consistency.

Platform at a glance

EXECUTIVE OVERVIEW

100K

Total Transactions

₹42.42

Avg Transaction

5.87%

Failure Rate

2.0%

Fraud Rate

Feature Phone

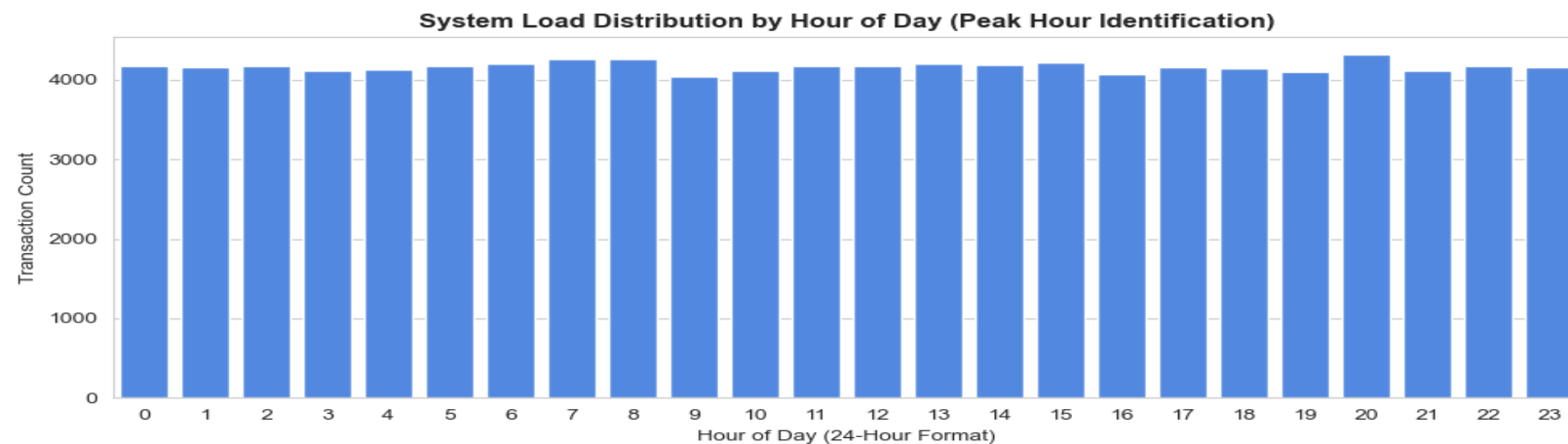
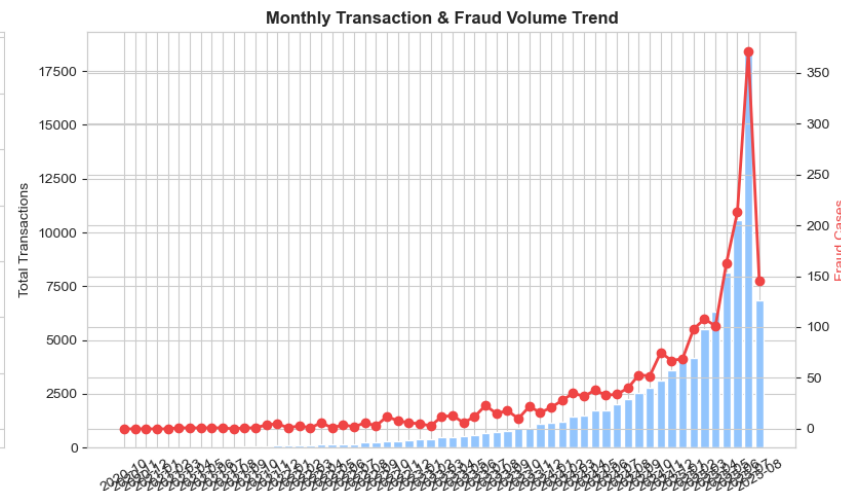
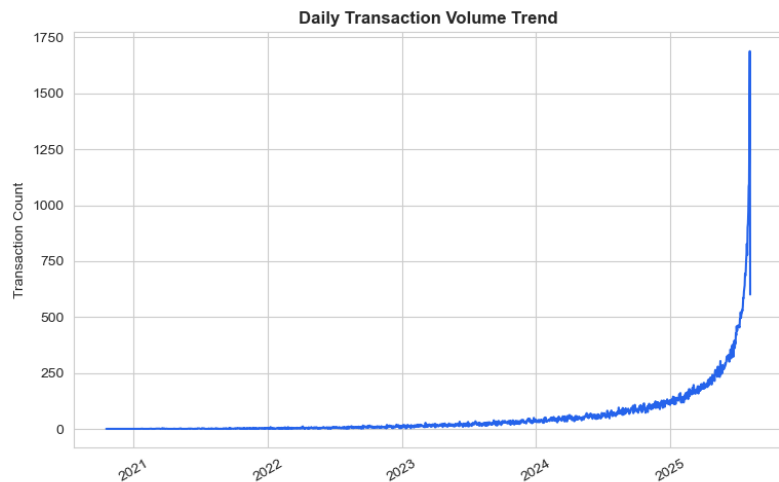
Top Device (2.2% fraud)

Apparel

Top Merchant (2.2% fraud)

₹42,41,774.26 total transaction value across 12 months.

Volume is compounding



WHAT WE SEE

4K → 21K

monthly volume, Sep to Aug

Peak windows

7–8 AM · 1–3 PM · 8–9 PM

~35% of daily volume

Off-peak

1–5 AM — safe
maintenance window

Device integrity is the signal

THE FRAUD SIGNAL

1.39%

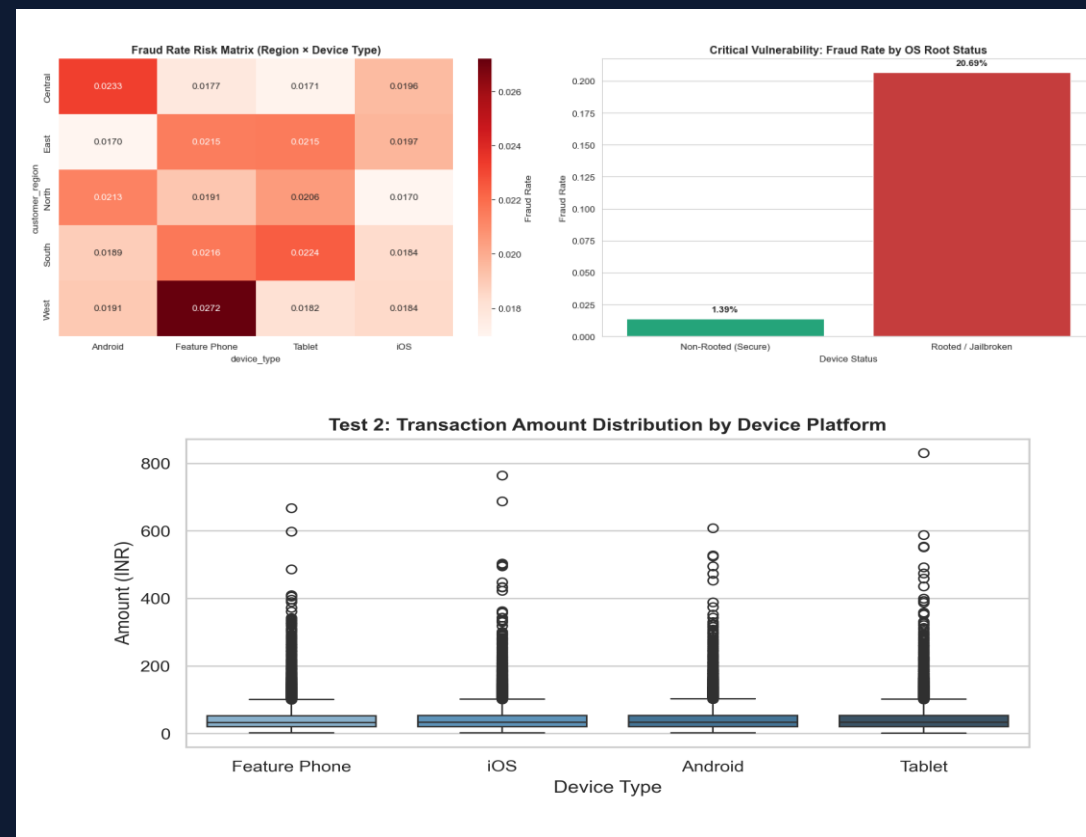
Secure devices

20.69%

Rooted / jailbroken

≈15× fraud risk · Chi-square $p \approx 0.000$ · Cramér's V 0.2414

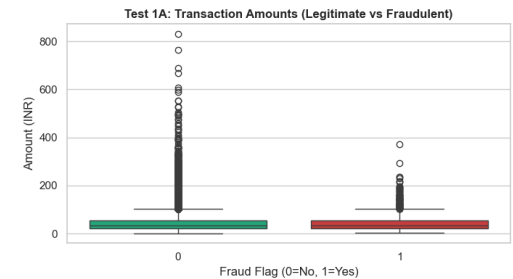
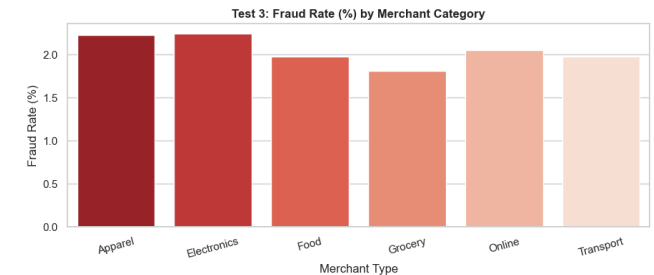
Device type barely moves fraud — root status does.



What matters — and what doesn't

HYPOTHESIS TEST	METHOD	EFFECT SIZE	VERDICT
Fraud rate by device root status	Chi-square	Cramér's V 0.2414 · medium	Significant
Transaction amount by fraud status	Welch t-test	Negligible	Not significant
Amount: North vs West region	Welch t-test	Negligible	Not significant
Amount across device types	ANOVA	Negligible	Not significant
Fraud rate by merchant type	Chi-square	Negligible	Not significant
Fraud rate by channel	Chi-square	Negligible	Not significant
Transaction status by device	Chi-square	Negligible	Not significant
Customer risk score vs fraud	Point-biserial	$r = -0.0016$	Not significant

Only device root status shows a real effect.



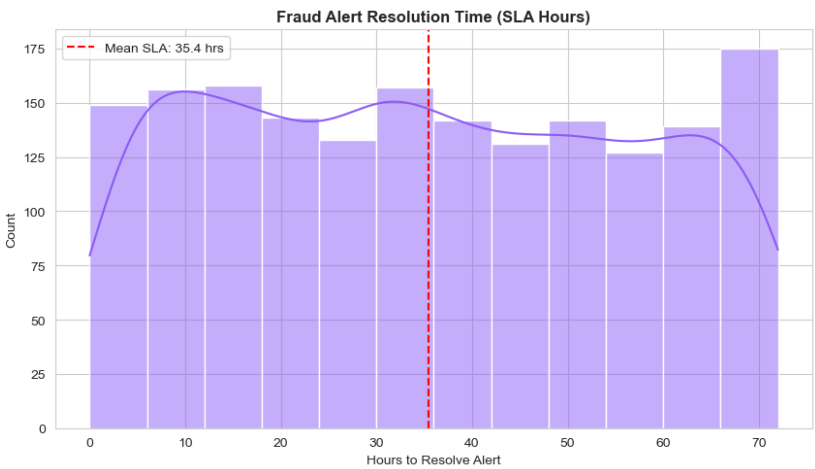
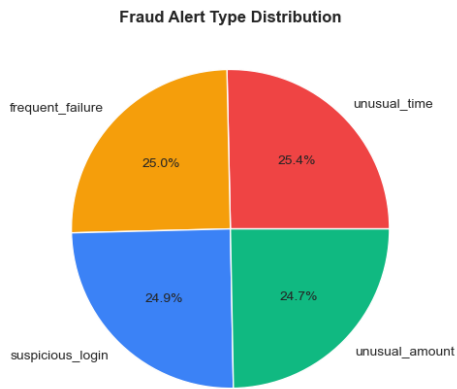
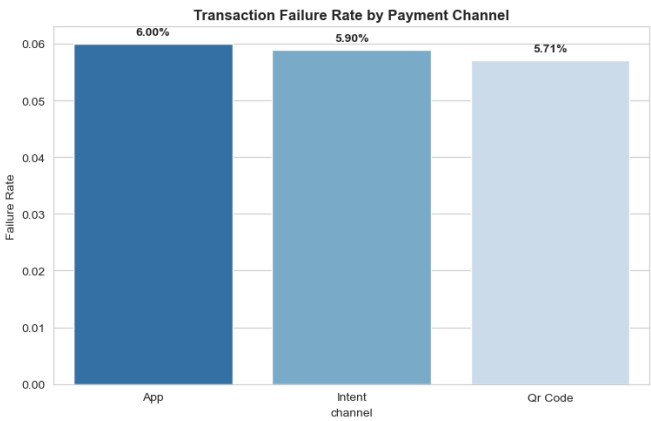
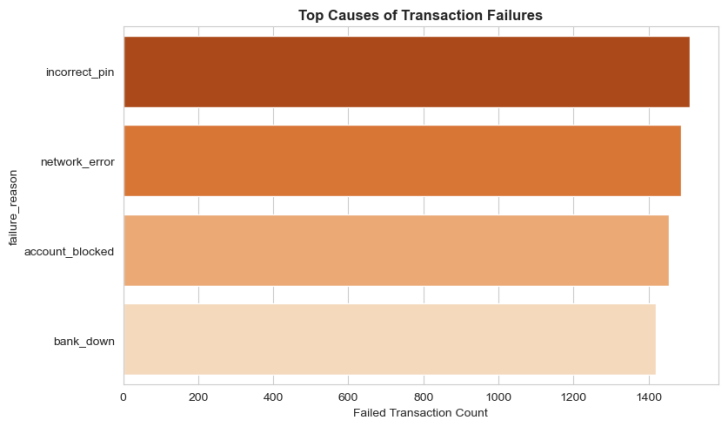
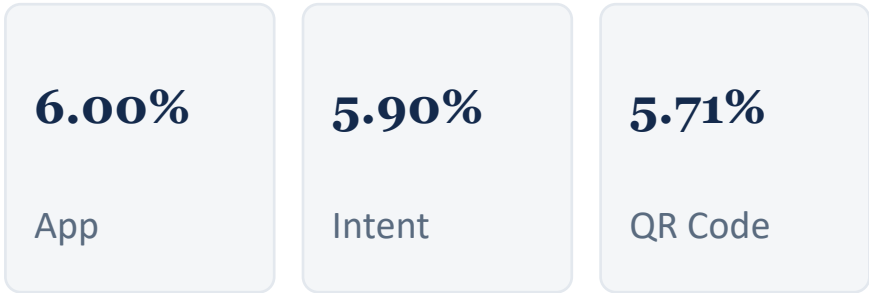
Where transactions break

TOP FAILURE REASONS (RANKED)

- 1 Incorrect PIN
- 2 Network Error
- 3 Account Blocked
- 4 Bank Down

Ordered by contribution to the 5.87% failure rate.

FAILURE RATE BY CHANNEL



FRAUD ALERT SLA: 87.6% resolved · 35.4 h avg resolution · 248 still open

Who transacts, who carries risk

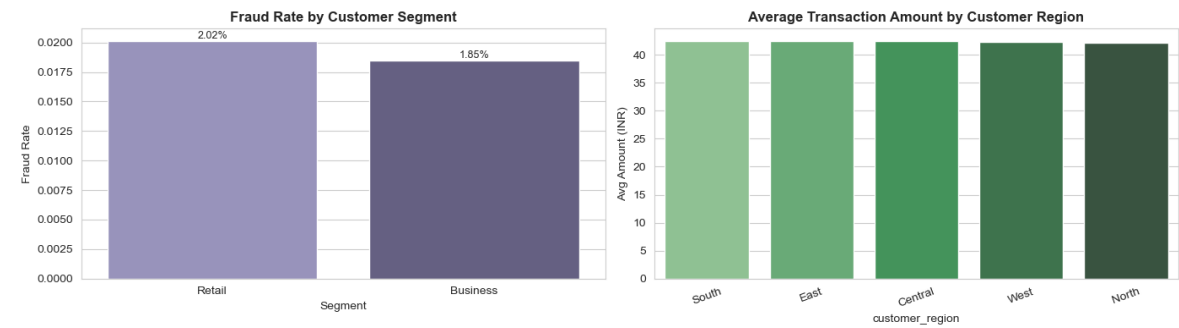
SEGMENT FRAUD RATE

2.02%

Retail

1.85%

Business



MERCHANT TYPES BY FRAUD RATE

Electronics



2.2%

Apparel



2.2%

Grocery



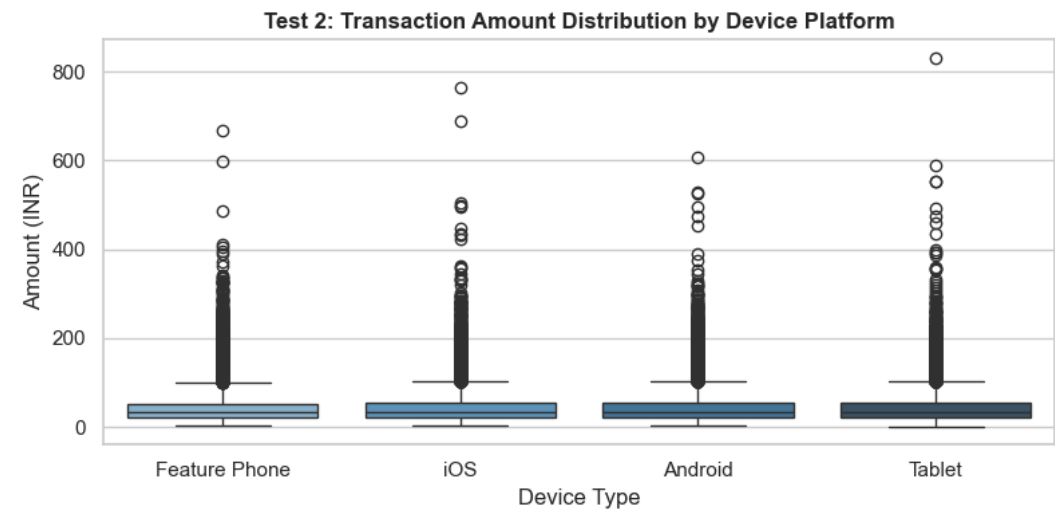
1.8%

7K

Active Customers

501

Active Merchants



Risk sits with device integrity, not customer segment.

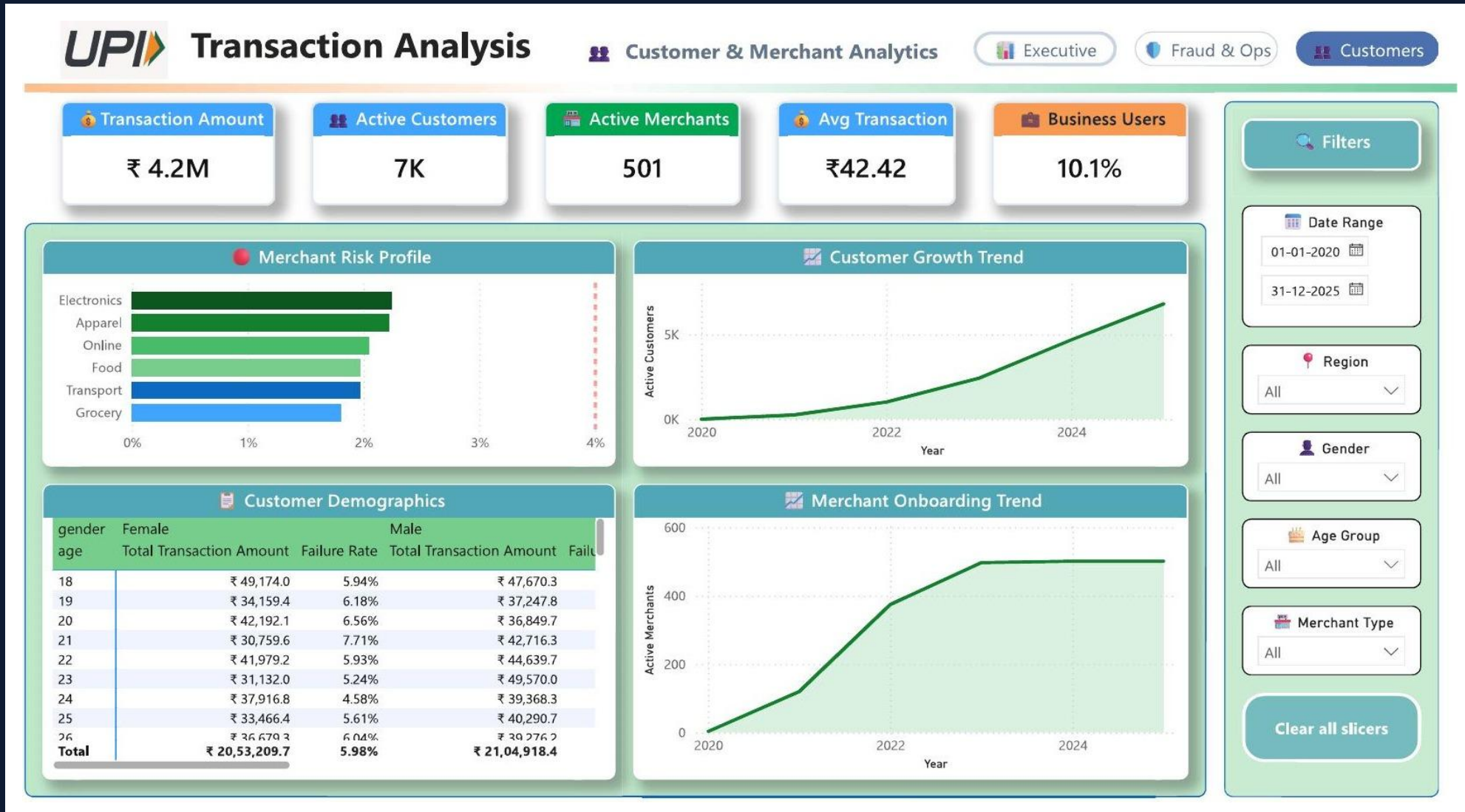
Executive Overview



Fraud & Ops



Customers & Merchants



Five moves that change the numbers

01

Block or step-up rooted devices

~15× fraud risk concentrated here

02

Automate 4-hour fraud triage

35.4 h avg resolution · 248 open

03

Biometric auth to cut PIN failures

Incorrect PIN is failure driver #1

04

Recalibrate risk score with device signals

Current score: $r = -0.0016$ vs fraud

05

Plan capacity around peak windows

7–8 AM · 1–3 PM · 8–9 PM

Thank you

Career 247 · Data Analytics with GenAI · UPI Transaction Analysis